

UCS301 Data Structures
Lab Assignment 5: Single Linked List
(Week 5)

1. Develop a menu driven program for the following operations on a Singly Linked List.
 - (a) Insertion at the beginning.
 - (b) Insertion at the end.
 - (c) Insertion in between (before or after a node having a specific value, say 'Insert a new Node 35 before/after the Node 30').
 - (d) Deletion from the beginning.
 - (e) Deletion from the end.
 - (f) Deletion of a specific node, say 'Delete Node 60').
 - (g) Search for a node and display its position from head.
 - (h) Display all the node values.
2. Write a program to count the number of occurrences of a given key in a singly linked list and then delete all the occurrences.

Input: Linked List : 1->2->1->2->1->3->1 , key: 1

Output: Count: 4 , Updated Linked List: 2->2->3.

3. Write a program to find the middle of a linked list.

Input: 1->2->3->4->5

Output: 3

4. Write a program to reverse a linked list.

Input: 1->2->3->4->NULL

Output: 4->3->2->1->NULL